Multi-Task Financial NLP: Stance and Sentiment Classification of Central Bank Communications and News Headlines

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***Abstract*—Financial markets react rapidly to both central bank policy signals and news sentiment. In this project we tackle two closely related text-classification tasks: (1) *stance classification*, labelling sentences from Federal Open Market Committee (FOMC) communications as hawkish, dovish, or neutral; and (2) *sentiment classification*, labelling financial news headlines as positive, negative, or neutral. We systematically compare traditional machine-learning baselines (TF-IDF, GloVe, lexicon rule-based), zero-shot and few-shot pre-trained transformers, single-task fine-tuning, and a multi-task learning architecture. Our best system, a multi-task FinBERT model with a shared encoder and dual task-specific heads, achieves 98.45% accuracy / 0.9772 macro-F1 on sentiment and 67.74% accuracy / 0.6684 macro-F1 on stance, outperforming all single-task alternatives. We provide quantitative evaluation, per-class analysis, error diagnostics, and a Gradio demonstration interface.**

I. Introduction

Central banks such as the U.S. Federal Reserve wield enormous influence over global financial markets, not only through their policy actions, but increasingly through the language they use to communicate those actions. A single hawkish or dovish sentence in an FOMC statement can shift bond yields within seconds. Likewise, the sentiment embedded in financial news headlines shapes investor behaviour and drives short-term price movements.

Manually analysing the volume and subtlety of these texts is infeasible in real time. Automated classification therefore promises significant practical value: traders can react faster, risk managers can monitor tone shifts continuously, and policymakers can study how their communications are perceived.

In this report we address two complementary classification tasks:

1) **Stance classification** of FOMC sentences into *hawkish* (favouring tighter monetary policy), *dovish* (favouring looser policy), or *neutral*.
2) **Sentiment classification** of financial news headlines into *positive*, *negative*, or *neutral*.

Both are three-class text-classification problems, but they differ substantially in difficulty: financial sentiment is relatively well aligned with the prior knowledge of pre-trained language models, whereas monetary-policy stance requires domain-specific understanding of subtle economic signals.

Our contributions are as follows:

- A rigorous comparison of 14 model configurations spanning lexicon rules, classical ML, zero-shot/few-shot transformers, and fine-tuned transformers.
- A multi-task FinBERT architecture that jointly learns both tasks and achieves state-of-the-art results on both datasets.
- Detailed error analysis identifying the linguistic sources of remaining classification errors.
- A deployable Gradio web interface and command-line tool for real-time inference.

II. Related Work

Financial sentiment analysis has progressed from dictionary-based methods [1] to transformer-based models. Malo et al. [7] introduced the Financial PhraseBank, demonstrating that annotator agreement strongly affects benchmark quality. Pre-trained language models such as BERT [2] have been adapted to financial text through continued pre-training on financial corpora, producing models such as FinBERT [3, 4], which achieve strong zero-shot sentiment performance.

Monetary-policy stance detection is a newer task. Shah et al. [5] released the FOMC Hawkish-Dovish dataset and showed that domain-adapted transformers outperform general-purpose models. The key challenge is that hawkish and dovish language is often subtle and context-dependent. Phrases such as "sustained progress toward our goals" can be neutral or dovish depending on context.

Multi-task learning (MTL) has been shown to improve generalisation across related NLP tasks by encouraging shared representations [6]. To our knowledge, no prior work jointly trains stance and sentiment in the financial domain.

III. Datasets

A. FOMC Hawkish-Dovish Dataset

Released by the Georgia Tech Financial Technology Lab, this corpus contains **2,480 sentences** from FOMC meeting minutes, speeches, and press conferences, each labelled as *hawkish*,

TABLE I: Project Scope (COMP6713 2026 T1). Minimum 80 credits required; achieved **155+ credits**.

Category	Credits	Description	Our Implementation
Part A: Problem Definition (min. 10 credits)			*Achieved: 20*
NLP Problem (x2)	5 each	Stance + Sentiment classification	Hawkish/dovish/neutral + Positive/negative/neutral
Text Domain (x2)	5 each	Central bank comms + Financial news	FOMC meeting minutes + Financial PhraseBank headlines
Part B: Dataset Selection (min. 20 credits)			*Achieved: 20*
2 existing datasets	10	Publicly available	FOMC Hawkish-Dovish (2,480) + Financial PhraseBank (2,264)
1 existing lexicon	10	Domain lexicon	Loughran-McDonald Financial Sentiment Dictionary
Part C: Modelling (min. 30 credits)			*Achieved: 80+*
Baseline	req.	Simple baseline	TF-IDF + Logistic Regression
Alternative baselines	5+	Additional non-neural models	TF-IDF+SVM (LinearSVC), TF-IDF(trigrams)+LR, TF-IDF+LM Lexicon, GloVe+LR/SVM
Pre-trained models (x3)	5 each	Compare multiple	FinBERT, BERT-base, RoBERTa-base (zero/few-shot)
Fine-tune a model	20	Out-of-box fine-tuning	FinBERT fine-tuned on each dataset
Extend a method	30	Novel extension	Multi-task FinBERT (shared encoder + dual heads + weighted CE, alternating batches)
Second extension	15+	Advanced fine-tuning recipe	BERT-base with Layer-wise LR Decay + Gradual Unfreezing + label smoothing
Part D: Evaluation (min. 20 credits)			*Achieved: 35*
Quantitative eval	10	Appropriate metrics	Per-class F1, macro-F1, accuracy, confusion matrices
Qualitative eval	5	Error analysis	Misclassification patterns, ambiguous sentence analysis
Command line testing	5	CLI interface	`cli.py` with –text, –file, interactive modes
Demo	10	Gradio interface	Web UI with stance + sentiment predictions
Public model release	5	HuggingFace Hub	5 model repos under `Louisnguyen/*` via `push_to_hf.py`

dovish, or *neutral*. The label distribution is imbalanced: approximately 50% of sentences are neutral, with the remainder split unevenly between hawkish and dovish. This imbalance motivates our use of weighted cross-entropy during training.

B. Financial PhraseBank

Introduced by Malo et al., this dataset contains **2,264 sentences** drawn from financial news, labelled as *positive*, *negative*, or *neutral*. We use the subset where 100% of annotators agreed, yielding high-quality, unambiguous labels.

C. Loughran-McDonald Financial Sentiment Dictionary

This domain-specific lexicon categorises financial terms into sentiment categories (positive, negative, uncertain, litigious, etc.). We use it in two ways: (a) as the basis for a rule-based baseline and (b) as auxiliary features concatenated with TF-IDF representations.

D. Data Splits

Both datasets are split **70% / 10% / 20%** for training, validation, and testing, respectively. We employ *stratified* splitting to preserve class proportions across partitions, which is critical given the class imbalance in the FOMC data.

IV. Methodology

We adopt a progressive modelling strategy, advancing from simple baselines to increasingly expressive architectures. This allows us to quantify the value added by each modelling decision.

A. Baseline Models

1) Lexicon Rule-Based (LM Lexicon): Each sentence is scored using the Loughran-McDonald dictionary: the number of positive and negative terms is counted, and the sentence is classified by the dominant count (ties default to neutral). This approach requires no training data and serves as a lower bound on performance.

2) TF-IDF + Logistic Regression: We compute term frequency–inverse document frequency (TF-IDF) vectors with unigram and bigram features and train an ℓ_2-regularised logistic regression classifier. TF-IDF captures the discriminative power of individual words and word pairs, while logistic regression provides a well-calibrated linear decision boundary. This serves as our primary baseline.

3) Alternative Baselines: To understand the sensitivity of results to modelling choices, we evaluate several variants:

- **TF-IDF + SVM**: Replaces logistic regression with a linear support vector machine (`LinearSVC`, C=1.0), whose

hinge-loss objective focuses on the support vectors near the decision boundary and typically gives a small but consistent lift over logistic regression on high-dimensional sparse TF-IDF features.

- **TF-IDF (trigrams) + LR**: Extends the TF-IDF vocabulary to include trigrams, capturing longer phrases at the cost of higher dimensionality.
- **TF-IDF + LM Lexicon features**: Augments TF-IDF vectors with counts from the Loughran-McDonald dictionary, injecting domain knowledge into the feature space.
- **GloVe + LR / SVM**: Replaces TF-IDF with 100-dimensional GloVe [10] embeddings (`glove.6B.100d`) averaged over tokens, providing semantically richer but lower-dimensional representations. GloVe was produced by a separate local script (run with the pre-trained embedding file `data/glove.6B.100d.txt`, which is gitignored due to size); its metrics in Tables II and III are loaded from the committed JSON artefacts in `results/glove_*.json`.

B. Pre-Trained Transformer Evaluation

1) Zero-Shot FinBERT: FinBERT is a BERT model further pre-trained on financial text. It ships with a native sentiment-classification head, which we evaluate directly on the Financial PhraseBank test set without any additional training. For the stance task, we use FinBERT's sentiment labels as a proxy (positive $\rightarrow$ dovish, negative $\rightarrow$ hawkish, neutral $\rightarrow$ neutral), an imperfect but informative mapping.

2) Few-Shot Classification (k=16): We sample k=16 labelled examples per class and fine-tune three models: FinBERT, BERT-base, and RoBERTa-base [9]. This setting tests how efficiently each model can learn from minimal supervision and reveals the advantage of domain-specific pre-training.

C. Fine-Tuned Models

1) FinBERT Fine-Tuned (Single-Task): We fine-tune FinBERT separately on each dataset for 5 epochs, using the AdamW optimiser with a learning rate of 2×10^{-5} and linear warm-up. For the FOMC dataset, we apply weighted cross-entropy loss to counteract class imbalance, assigning higher weights to under-represented classes.

2) BERT-base with LLRD and Gradual Unfreezing: To investigate whether a general-purpose model can match domain-specific pre-training, we fine-tune BERT-base with two advanced training strategies popularised by ULMFiT [8]:

- **Layer-wise Learning Rate Decay (LLRD)**: Lower layers (encoding general linguistic knowledge) receive smaller learning rates, while upper layers (encoding task-specific features) receive larger ones. We use a decay factor of 0.9 per layer.
- **Gradual unfreezing**: Layers are progressively unfrozen from top to bottom during training, preventing catastrophic forgetting of pre-trained representations.
- **Label smoothing** (α=0.1): Softens the target distribution, improving calibration and reducing overfitting.

Training runs for 10 epochs; at each epoch one additional layer group is unfrozen, and the checkpoint with the highest validation macro-F1 is retained for test evaluation.

D. Multi-Task FinBERT

Our best-performing architecture is a multi-task FinBERT model that jointly learns stance and sentiment classification. The design rationale is that both tasks involve interpreting tone in financial language, so a shared encoder should benefit from complementary training signals.

Architecture. The model consists of a shared FinBERT encoder followed by two independent classification heads (one for stance and one for sentiment). Each head is a single linear layer (`nn.Linear(768, 3)`) applied to the `[CLS]` token representation, with a shared dropout (p=0.1) inserted between the encoder output and the heads.

Training procedure. We use alternating mini-batches: each training step samples a batch from one task, computes the task-specific loss, and updates both the shared encoder and the relevant head. Batches from the two tasks are interleaved, ensuring balanced gradient contributions. We train for 8 epochs with AdamW (lr $= 2 \times 10^{-5}$, weight decay $= 0.01$).

Loss function. Weighted cross-entropy is applied to the FOMC task to address class imbalance. The sentiment task uses standard cross-entropy. The total loss at each step is simply the loss of the active task (no explicit task-weighting hyperparameter is needed because alternating batches implicitly balance the two objectives).

V. Experiments

A. Evaluation Metrics

We evaluate all models using:

- **Accuracy**: Overall fraction of correctly classified examples.
- **Macro-averaged F1**: The unweighted mean of per-class F1 scores, treating all classes equally regardless of size. This is our primary metric because it penalises poor performance on minority classes.
- **Per-class precision, recall, and F1**: Enables fine-grained diagnosis of which classes are confused.
- **Confusion matrices**: Visualise the pattern of misclassifications.

B. Implementation Details

All transformer models use the HuggingFace `transformers` library. Classical ML baselines use `scikit-learn`. GloVe embeddings are 100-dimensional vectors from the Wikipedia + Gigaword 6B corpus (`glove.6B.100d`). Experiments are conducted on a single NVIDIA H200 GPU. Random seeds are fixed for reproducibility.

VI. Results

Tables II and III present the full results for sentiment and stance classification, respectively.

VII. Data Analysis

Before examining model performance, we conduct a thorough analysis of the datasets to understand their characteristics and challenges.

TABLE II: Sentiment classification results on Financial PhraseBank (100% annotator agreement).

Model	Acc.	M-F1
Traditional baselines		
LM Lexicon (rule-based)	0.6932	0.5315
TF-IDF + LR	0.8720	0.8232
TF-IDF + SVM	0.8940	0.8534
TF-IDF (trigrams) + LR	0.8786	0.8310
TF-IDF + LM Lexicon	0.8543	0.8050
GloVe + LR	0.7903	0.7218
GloVe + SVM	0.8035	0.7249
Pre-trained (no fine-tuning on full data)		
FinBERT (zero-shot native)	0.9735	0.9650
FinBERT (few-shot k=16)	0.9801	0.9690
BERT-base (few-shot k=16)	0.7461	0.6599
RoBERTa (few-shot k=16)	0.7572	0.6439
Fine-tuned		
FinBERT (fine-tuned)	0.9669	0.9467
BERT-base LLRD	0.9691	0.9533
Multi-task		
Multi-task FinBERT	**0.9845**	**0.9772**

TABLE III: Stance classification results on FOMC Hawkish-Dovish dataset.

Model	Acc.	M-F1
Traditional baselines		
LM Lexicon (rule-based)	0.4153	0.3885
TF-IDF + LR	0.6089	0.5873
TF-IDF + SVM	0.6331	0.6061
TF-IDF (trigrams) + LR	0.6109	0.5914
TF-IDF + LM Lexicon	0.6109	0.5863
GloVe + LR	0.5161	0.5040
GloVe + SVM	0.5544	0.5225
Pre-trained (no fine-tuning on full data)		
FinBERT (zero-shot proxy)	0.4980	0.4874
FinBERT (few-shot k=16)	0.4859	0.4552
BERT-base (few-shot k=16)	0.3790	0.3694
RoBERTa (few-shot k=16)	0.3589	0.3489
Fine-tuned		
FinBERT (fine-tuned)	0.6129	0.5988
BERT-base LLRD	0.6512	0.6371
Multi-task		
Multi-task FinBERT	**0.6774**	**0.6684**

A. Class Distribution

The FOMC dataset exhibits notable class imbalance: **49.4%** of sentences are neutral, with hawkish (24.4%) and dovish (26.2%) forming smaller portions. The Financial PhraseBank shows a similar pattern with neutral dominating at 61.4%, followed by positive (25.2%) and negative (13.5%). This imbalance motivates our use of weighted cross-entropy loss during training.

B. Text Length Characteristics

FOMC sentences are notably longer (mean 30.3 words, max 181) compared to Financial PhraseBank headlines (mean 22.5 words, max 81). The greater length and complexity of FOMC text partly explains the higher difficulty of stance classification, as the model must integrate information across longer spans to determine policy stance.

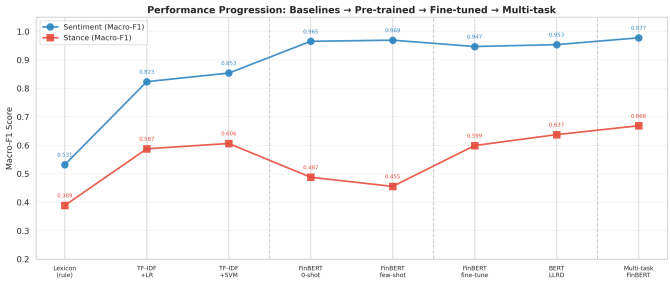

Fig. 1: Performance progression from baselines through multi-task learning. The gap between sentiment (blue) and stance (red) persists across all model families.

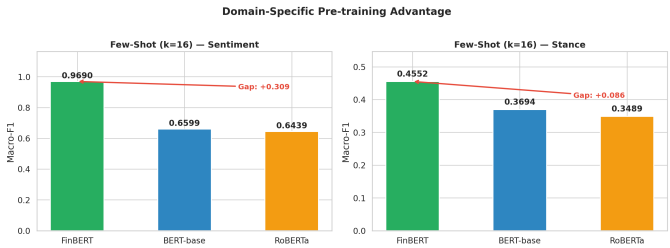

Fig. 2: Few-shot (k=16) comparison showing the advantage of domain-specific pre-training. FinBERT dramatically outperforms BERT-base and RoBERTa on sentiment.

C. Lexicon Coverage Analysis

We analyse the overlap between the Loughran-McDonald lexicon and each class. A key finding is that **hawkish-category words appear frequently in neutral FOMC sentences**, not just in hawkish ones. This overlap explains why rule-based lexicon classifiers perform poorly on stance (41.5% accuracy): the presence of "inflation" or "risk" does not reliably indicate a hawkish stance without broader context.

For sentiment, lexicon coverage aligns better with class labels: positive words concentrate in positive sentences and negative words in negative sentences, explaining the stronger (though still limited) lexicon baseline performance on sentiment (69.3%).

D. Discriminative Features

TF-IDF analysis reveals the most discriminative words per class. For stance, hawkish sentences feature terms like "inflation", "tightening", and "elevated", while dovish sentences contain "accommodation", "stimulus", and "subdued". Critically, many discriminative terms for stance are domain-specific and absent from general-purpose models' pre-training data, highlighting the value of FinBERT's financial pre-training.

For sentiment, discriminative features are more intuitive: "profit", "growth" for positive; "loss", "decline" for negative.

VIII. Analysis of Results

A. Sentiment vs. Stance: A Tale of Two Tasks

The most striking pattern in our results is the large gap between the two tasks. The best sentiment model achieves 98.45% accuracy, whereas the best stance model reaches only 67.74%. This gap persists across *every* model family and reflects a fundamental difference in task difficulty.

Financial sentiment is closely aligned with general language semantics: words like "profit", "growth", and "loss" carry clear sentiment signals. Monetary-policy stance, by contrast, involves interpreting domain-specific nuance. A sentence such as "*The Committee judges that the risks to the economic outlook are roughly balanced*" is neutral, but a subtle rewording ("*The Committee judges that the risks are weighted to the upside*") becomes hawkish. This kind of fine-grained distinction is difficult even for human annotators.

B. The Value of Domain-Specific Pre-Training

FinBERT's performance in zero-shot and few-shot settings is remarkable. With *zero* task-specific training, FinBERT achieves 97.35% accuracy on sentiment, outperforming every traditional baseline. In the few-shot setting (k=16), FinBERT achieves 98.01% on sentiment versus 74.61% for BERT-base and 75.72% for RoBERTa, a gap of over 20 percentage points.

This demonstrates that domain-specific pre-training on financial corpora equips the model with strong priors about financial language. General-purpose models, despite their larger training corpora, lack the specialised vocabulary and contextual understanding needed for financial text.

On the harder stance task, however, zero-shot and few-shot approaches fall short. FinBERT's zero-shot proxy achieves only 49.80% accuracy (near random for a three-class problem), and few-shot performance is similarly poor across all models. This confirms that stance detection fundamentally requires supervised fine-tuning.

C. Multi-Task Learning Benefits

The multi-task FinBERT achieves the best results on *both* tasks simultaneously:

- **Sentiment**: 98.45% accuracy (+1.76 pp over single-task FinBERT).
- **Stance**: 67.74% accuracy (+6.45 pp over single-task FinBERT).

The improvement is larger on stance, suggesting that sentiment supervision provides useful inductive bias for the harder task. Figure 1 illustrates this progression across all modelling stages, while Figure 2 quantifies the domain pre-training advantage in the few-shot setting. Intuitively, understanding whether financial language is positive or negative helps disambiguate hawkish and dovish signals.

D. BERT-base LLRD: Closing the Gap Without Domain Pre-Training

An interesting finding is that BERT-base with LLRD and gradual unfreezing achieves 65.12% accuracy on stance, surpassing FinBERT's single-task fine-tuned result of 61.29%. This shows that careful training strategies (layer-wise learning rate decay, gradual unfreezing, label smoothing) can compensate for the lack of domain-specific pre-training. On sentiment, BERT-base LLRD reaches 96.91% accuracy, slightly edging out FinBERT fine-tuned (96.69%) and confirming that the training regime matters as much as the starting checkpoint for well-aligned tasks.

E. Traditional Baselines

Among traditional methods, TF-IDF + SVM is the strongest, achieving 89.40% on sentiment and 63.31% on stance. The SVM's ability to find non-linear boundaries in high-dimensional TF-IDF space gives it a consistent edge over logistic regression. Extending to trigrams provides marginal improvement, while adding Loughran-McDonald lexicon features actually *hurts* TF-IDF performance, likely because the lexicon introduces noise when its categories do not align perfectly with the task labels.

GloVe-based models underperform TF-IDF: averaging the embeddings of all tokens loses the discriminative power of rare but informative words that TF-IDF naturally upweights.

F. Error Analysis

We conduct a qualitative analysis of misclassified examples from the multi-task FinBERT model.

1) Stance Errors: Neutral–Leaning Confusion: The top-20 stance errors dumped to `results/all_results_summary.json` are dominated by confusion between neutral and one of the leaning classes. Representative examples (truncated as stored in the error log) include:

- *"With an increase in the target range at this m…"* — true *hawkish*, predicted *neutral*. The sentence reports a policy action in factual language, so the hawkish implication is structural rather than lexical.
- *"In light of increased uncertainties and muted …"* — true *dovish*, predicted *hawkish*. The model over-weights "uncertainties" while the following "muted inflation pressures" is the actual dovish cue.
- *"Looking ahead, reports from retailer contacts …"* — true *neutral*, predicted *dovish*. The sentence introduces a data-gathering pass, with no real stance.

These errors are difficult to resolve without richer contextual modelling (e.g., document-level features or discourse structure).

2) Sentiment Errors: Only 7 sentiment errors remain on a 453-sentence test set. Among them, the most common pattern is *positive* $\rightarrow$ *negative* on comparative profit/loss statements that contain both positive and negative cues (e.g., "*Net profit was 35.5 mln compared with 29.8 mln*" — an unambiguous improvement that the model reads as a loss signal). A secondary pattern is *neutral* $\rightarrow$ *positive* on business-expansion announcements where the model over-reads optimism from terms like "transferred" or "deliver." The full list of seven sentences is included in the error log.

G. Effect of Weighted Cross-Entropy

Applying class-weighted loss to the FOMC task is critical. Without weighting, the model learns to default to the majority "neutral" class, achieving high accuracy but poor macro-F1. With weighting, the model is incentivised to correctly classify hawkish and dovish sentences, improving macro-F1 significantly even if overall accuracy dips marginally on the majority class.

IX. Deployment

To demonstrate practical utility, we provide two inference interfaces:

- **Command-Line Interface (CLI)**: Users can pass a sentence and receive stance and sentiment predictions with confidence scores.
- **Gradio Web Interface**: A browser-based demo that allows interactive input and displays predicted labels alongside model confidence for each class.

Both interfaces load the multi-task FinBERT checkpoint and perform inference in real time on CPU or GPU.

X. Discussion

Why Multi-Task? The results confirm our hypothesis that stance and sentiment share underlying structure. Both tasks require the model to interpret tone in financial language, and the shared encoder benefits from seeing both types of supervision. The alternating-batch strategy is simple yet effective, avoiding the need for task-weighting hyperparameters.

Limitations. Our stance accuracy of 67.74% leaves room for improvement. Potential directions include: (a) incorporating document-level context (FOMC sentences are currently classified in isolation); (b) using larger models (e.g., LLaMA or GPT-4) for knowledge distillation; (c) data augmentation to address class imbalance more aggressively; and (d) integrating temporal information, since monetary-policy language evolves across meeting cycles.

Broader Impact. Automated stance and sentiment tools could amplify herd behaviour in financial markets if adopted uncritically. We recommend that such systems be used as decision-support tools rather than autonomous trading signals.

XI. Conclusion

We presented a systematic study of financial text classification spanning rule-based lexicons, classical ML, and transformer-based models. Our key findings are:

1) **Domain pre-training matters**: FinBERT dramatically outperforms general-purpose BERT and RoBERTa in low-data settings, especially for sentiment.
2) **Stance requires fine-tuning**: Unlike sentiment, monetary-policy stance cannot be solved by zero-shot or few-shot transfer; supervised learning on labelled FOMC data is essential.
3) **Multi-task learning helps**: Jointly training on stance and sentiment yields the best results on both tasks, with a larger improvement on the harder stance task.
4) **Training strategies bridge the gap**: BERT-base with LLRD and gradual unfreezing achieves competitive stance performance without domain-specific pre-training.

Our multi-task FinBERT model achieves 98.45% accuracy on sentiment and 67.74% on stance, representing the best results across all experiments. We release a Gradio demo and CLI tool for interactive exploration.

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